ADA LOVELACE

1815–1852

The mathematician and writer Ada Lovelace saw the potential of Babbage's calculating machine. Her instructions for it, written in algebraic code, are regarded as the first-ever computer program.

Born Ada Byron, Lovelace was the only legitimate child of the poet Lord Byron, who was briefly married to her mother, Anne Isabella Milbanke. Aged 17, Ada's tutor, Mary Somerville, introduced her to Charles Babbage. Sharing a passion for mathematics and machines, the two became friends. In 1842, Babbage asked Lovelace to translate a paper about his Analytical Engine, written by the Italian engineer Luigi Menabrea. She added explanatory footnotes to the paper, tripling its length. In these, she set out how to code formulae as instructions for the calculating machine. Lovelace realized that if machines could manipulate numbers, they could also manipulate symbols and process algorithms. Her notes inspired Alan Turing's work on modern computers in the 1940s.

MILESTONES

FLYING ENGINE
In 1828, aged 12, she decides she wants to fly and designs a steam-powered flying machine.

EARLY INFLUENCES
Learns mathematics from her tutor, Mary Somerville, who in 1833 introduces her to Babbage.

A KEEN INTELLECT
Presented at Court in 1833; her brilliance is recognized. She starts to correspond with scientists.

COMPUTER PROGRAM
"Sketch of the Analytical Engine ... with Notes by the Translator" is published in 1843, with her own code.

ANNUAL MEMORIAL
Commemorated on the yearly Ada Lovelace Day from 2009, which celebrates women in science.